

AUTOMATIC RAILWAY GATE CONTROL SYSTEM USING IR SENSOR

PHASE 4 — PROTOTYPE DEVELOPMENT REVIEW (PDR)

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Phase duration: 22 August – 04 September 2026

1. Phase 4 Objective

Phase 4 converts the approved Phase 3 system design into a laboratory-scale Working Prototype (Version 1). The implementation follows the Expo guideline deliverables: working prototype, embedded software, PCB fabrication status where applicable, mechanical assembly, hardware integration, software integration, GitHub repository, and weekly progress report. The guideline also requires submission of the GitHub repository, prototype images, prototype video, source code, and weekly progress report, with a 3–5 minute prototype video.

2. Prototype Version 1 Overview

The prototype implements the dual-IR-sensor railway crossing concept developed in Phases 1–3. An approach-side IR sensor and a departure-side IR sensor provide digital detection signals to an Arduino Uno. The controller processes the sensor inputs and controls an SG90 servo-driven barrier together with an LED and buzzer warning system. Serial communication through the Arduino USB connection is used for event logging during development and validation.

3. Prototype Hardware

The hardware selected in Phase 3 is carried forward without changing the core architecture. The prototype uses pre-built sensor and actuator modules connected through a breadboard/module-level wiring arrangement.

- **Arduino Uno (ATmega328P):** central controller responsible for sensor reading, decision logic, timing, and actuator commands.
- **IR Sensor 1 – Approach:** detects the approaching train/object before the crossing.
- **IR Sensor 2 – Departure:** detects train/object clearance after the crossing.
- **SG90 Servo Motor:** drives the lightweight prototype barrier arm.
- **5 mm LED + 220 Ω resistor:** provides visual warning/status indication.
- **5 V Piezo Buzzer:** provides audible warning.
- **5 V supply / USB:** supplies the controller and prototype modules; all grounds are common.

4. Pin Connection Table

Component / Signal	Arduino Pin	Direction	Function
IR Sensor 1 OUT (Approach)	D2	Input	Digital beam-break input
IR Sensor 2 OUT (Departure)	D3	Input	Digital beam-break input
Servo Signal	D9	Output (PWM)	Barrier position control
LED Anode (through 220 Ω)	D7	Output	Visual alert
Buzzer (+)	D8	Output	Audible alert
IR VCC / Servo V+	5 V rail	Power	Shared supply
All module grounds	GND rail	Power	Common ground

The Phase 3 design specifies HIGH for an intact IR beam and LOW for a broken beam. The sensor modules, servo, LED and buzzer share the common supply/ground arrangement defined in the system design.

5. Mechanical Assembly

- Mount the two IR sensor modules securely on opposite sides of the model crossing so that their sensing paths are aligned with the intended train/object path.
- Fix the SG90 servo firmly to the model base and attach the lightweight barrier arm to the servo horn.
- Position the barrier so that its movement is clear of the track/model and does not obstruct the sensor beam.
- Keep wiring secured and insulated; avoid exposed or loose connections during demonstration.

6. Embedded Software

The embedded software is developed using the Arduino IDE. The firmware reads the two digital IR inputs, applies the planned validation/debounce logic, controls the gate state, operates the LED and buzzer, and sends event information through the Serial Monitor for development and validation. The Phase 3 architecture defines the gate states as IDLE, ALERTING, CLOSING, CLOSED and OPENING.

7. Software Integration Sequence

- Initialize the IR sensor inputs, servo output, LED and buzzer.
- Continuously read the approach and departure sensor signals.
- Validate sensor changes using the configured debounce/validation period.
- When a valid approach event is detected, activate the warning indication and command the gate toward the closed position.
- Keep the gate in the required closed/occupied state while the crossing is considered occupied.
- After a valid departure event, command the servo toward the open position and return the system to the normal state.
- Record relevant detection and gate events through the Serial Monitor during development.

8. Hardware Integration

Hardware integration combines the sensing, control, actuation and alert layers into one prototype. The two IR outputs are connected to Arduino digital inputs D2 and D3. The Arduino drives the servo through D9, the LED through D7 using a 220 Ω series resistor, and the buzzer through D8. The common 5 V and GND rails provide the required shared reference for the modules.

9. Software Integration

The firmware integrates sensor acquisition, debounce/validation, gate-state decision logic, alert outputs, servo control and serial event logging. The logging path is treated as a support function for development and testing and is not intended to replace the real-time control path.

10. Prototype Working Sequence

Step	System action
1	System starts in its defined normal state and initializes all modules.
2	Approach IR sensor detects a valid beam interruption.
3	Controller validates the event and activates the warning indication.
4	Servo receives the gate-position command and moves the prototype barrier.
5	The system maintains the gate state while the crossing remains occupied according to the sensor logic.

Step	System action
6	Departure IR sensor detects clearance.
7	Controller validates the departure event and commands the barrier to reopen.
8	System returns to the normal monitoring state and continues sensing.

11. PCB Fabrication Status

PCB fabrication is not applicable to the present Version 1 breadboard/module-based prototype. Phase 3 explicitly selected module-level point-to-point wiring for this prototype stage. A dedicated PCB can be considered in a later refinement if required.

12. GitHub and Version Control

The project follows the Git/GitHub workflow specified by the Expo guideline. The repository should contain the Phase 4 source code, circuit documentation, progress records and relevant prototype evidence. The guideline specifies one fork per team, one branch per team member, Team Lead review of pull requests, and faculty/mentor review before final integration.

GitHub Repository: [Insert actual repository link here]

Team branches: [Insert branch names if required]

13. Weekly Progress Report

Week / Date	Development activity	Evidence / status
Week 5 22–28 Aug	Prototype component arrangement; Arduino, IR sensors, servo, LED and buzzer integration; progress photos and video.	Attach integration progress photos and video.
Week 6 29 Aug–04 Sep	Firmware integration, sensor-to-servo control, alert integration, mechanical assembly, progress photos and video.	Attach assembly progress photos and video.

14. Prototype Evidence to be Submitted

- **Prototype Images:** Attach clear photographs showing the complete Version 1 prototype, sensor positions, Arduino wiring, servo barrier and alert devices.
- **Prototype Video:** Prepare a 3–5 minute video demonstrating the actual prototype and explaining the detection, control and gate operation.
- **Source Code:** Attach the actual Arduino IDE source code used for Version 1.
- **GitHub Repository:** Insert the team's actual repository link.
- **Weekly Progress Report:** Attach dated development records and photographs.

15. Safety and Documentation Checklist

- Check all 5 V and GND connections before powering the prototype.
- Keep wiring organized and insulated during demonstration.
- Do not overload the selected power supply.
- Use the SG90 only for the lightweight prototype barrier for which it was selected.
- Maintain actual photographs, source code and progress records; do not fabricate testing evidence or demonstration results.

16. Phase 4 Outcome

Phase 4 establishes the implementation stage of the Automatic Railway Gate Control System Using IR Sensor. The Version 1 prototype brings together the dual-IR sensing layer, Arduino Uno control layer, servo-driven barrier, LED/buzzer alerting and serial logging path defined in Phase 3. The completed Phase 4 submission package consists

of the working prototype evidence, embedded source code, integration documentation, GitHub repository information and weekly progress report.

17. Conclusion

The Phase 4 Prototype Development Review translates the approved Phase 1 problem definition, Phase 2 concept and Phase 3 system design into a laboratory-scale implementation. The next project stage is Phase 5 — Testing & Validation, where functional test cases, performance evaluation, error analysis, debugging, power analysis and reliability testing will be documented using actual measured observations.

Reference basis: Project Expo-2026 Guidelines; Phase 1 — Problem Identification & Research; Phase 2 — Ideation & Concept Development; Phase 3 — System Design Review.